

Hypothalamic deiodinase type-3 establishes the period of circannual interval timing in mammals

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This study provides **important** findings on the understanding of circannual timing in mammals, for which iodothyronine deiodinases (DIOs) have been suggested to be of critical importance, yet functional genetic evidence has been missing. The authors **convincingly** implicate *dio3*, the major inactivator of the biologically active thyroid hormone T3, in circannual timing in Djungarian hamsters, using a combination of correlative and gene knock-out experiments; thus this provides key insights into the evolution and function of animal annual timing mechanisms.

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Abstract

Animals respond to environmental cues to time phenological events, but the intrinsic mechanism of circannual timing remains elusive. We used transcriptomic sequencing and frequent sampling of multiple hypothalamic nuclei in Djungarian hamster to examine the neural and molecular architecture of circannual interval timing. Our study identified three distinct phases of transcript changes, with deiodinase type-3 (*Dio3*) expression activated during the early induction phase. Subsequent work demonstrated that targeted mutation of *Dio3* using CRISPR-Cas resulted in a shorter period for circannual interval timing. Hamsters that are non-responsive to short photoperiod and fail to show any winter adaptations do not display changes in *Dio3* expression do not show any change in body mass or pelage. Our work demonstrates that changes in *Dio3* induction is essential for setting the period of circannual interval timing.

Introduction

Phenology of key life history traits is common across plant (Piao et al., 2019) and animal kingdoms (Cohen et al., 2018). The annual changes in daylength are the predominant environmental cue which animals use to time seasonal life history transitions (Liddle et al., 2022; Pérez et al., 2019; Wood and Loudon, 2014). Plants and animals also exhibit endogenous circannual timing in the absence of any change in environmental cues. For example, bird migration (Gwinner and Dittami, 1990), mammalian hibernation (Pengelley and Fisher, 1957), and reproduction (Woodfill et al., 1994; Lincoln et al., 2006) are all driven by robust intrinsically generated *circannual clocks*, the cycle of which nearly match a 12-month period. Some *circannual timers* estimate an interval period (e.g., 6-month) in which programmed changes in physiology and morphology occur in anticipation of the next season. Such interval timers are commonly observed as flowering in plants (Duncan et al., 2015), diapause in insects (Denlinger, 1974) and spring emergence in rodents (Prendergast et al., 2004). Interval timers are typically characterized by having light-dependent induction, maintenance and recovery phases.

Annual mammalian life history transitions are typically associated with major changes in energy demand (Ricklefs, 1991). While the melanocortin system, including neuropeptide y (*Npy*), agouti-related peptide (*Agrp*), and melanocortin receptors are known to regulate short-term changes in energy homeostasis (Yeo et al., 2021), the mechanisms and anatomical structures implicated in long-term, circannual variation in energy rheostasis are not well characterized. Somatostatin (*Sst*) (Marshall et al., 2024; Petri et al., 2016, 2014), proopiomelanocortin (POMC) (Bao et al., 2019; Mercer et al., 2000), and VGF nerve growth factor (Barrett et al., 2005) are known to be strong correlates of seasonal variation in energetic state. Tanycyte somas are essential for the integration of environmental and physiological signals required for circannual interval timing. Tanycytes are localized in the ependymal layer of the third ventricle and are highly sensitive to nutrient state (Bolborea et al., 2020) and receive photoperiodic signaling derived from thyrotropes in the pars tuberalis (Hanon et al., 2008; Wood et al., 2020). Previous work has established deiodinase type-2 (*Dio2*) and type-3 (*Dio3*) expression is anatomically localized to tanycytes and coordinate triiodothyronine-dependent annual transitions in physiological state (Bao et al., 2019; Ebling and Lewis, 2018; Hanon et al., 2008; Murphy et al., 2012; Petri et al., 2016; Wood et al., 2020). Here we delineate the molecular architecture of circannual interval timing by the hypothalamus and pituitary gland and tested the conjecture that *Dio3*, in the Djungarian hamster (*Phodopus sungorus*), is upregulated during the induction of circannual interval timing for energy rheostasis, and functions to establish the duration (or period) of the circannual interval timer.

Results

Male Djungarian hamsters were either kept in a long photoperiod (LP) control condition or moved from LP to short photoperiod (SP) conditions (Fig. 1). Djungarian hamsters remain in LP phenotype unless exposed to SP. Pelage color, torpor and body mass were used to monitor the induction, maintenance, and recovery phases of the circannual timer (Fig. 1A and C SI Appendix Movies S1 and 2). A full white pelage color was observed between 12-16 weeks after exposure to SP and gradually reversed to the LP agouti color after 28 weeks in SP. Hamsters engaged in torpor between 12-20 weeks in SP (movie S2). Massive, programmed changes in energy rheostasis were observed with a 30% reduction in daily food intake and a 20% decrease in average body mass by 12 weeks SP (Fig. 1C and D). Both food intake and body mass started to reverse to LP conditions after 20 weeks in SP. Epididymal adipose tissue mass and plasma insulin concentrations paralleled the change in body mass (Fig. 1D and SI Appendix Fig. S1). These reversals in

physiological state are indicative of the recovery phase of the circannual interval timer. GLP-1 did not display any change in circulating concentrations (**Fig. 1E**). Plasma glucose concentration jumped sharply after 16 weeks in SP (**Fig. 1E**), concurrent with the development of torpor in these animals (movie S2). The observed change in body mass, food intake, adipose tissue and plasma insulin versus the lack of change in GLP-1 highlights the physiological distinction between a programmed rheostatic mechanism characteristic of the circannual interval timing of energy stability in the former, while the latter are driven by short-term homeostatic mechanisms (Stevenson, 2024).

We then used Oxford Nanopore transcriptomic sequencing to characterize the molecular changes associated with phases of the circannual interval timer within multiple individual hypothalamic nuclei and the pituitary gland (**Fig. 1F** to G; fig. *SI Appendix S2* to 6 and Dataset S1 to 4). The development of refractoriness to melatonin, indicative of the recovery phase of the circannual timer, has been shown to develop independently in different hypothalamic regions (Freeman and Zucker, 2001). Molecular data was collected from the mediobasal hypothalamus (Bolborea et al., 2020), dorsomedial hypothalamus (Ebling et al., 2008), and the paraventricular nucleus (Bittman et al., 1991), along with the pituitary gland (Majumdar et al., 2023). This experimental approach provides a high-throughput and high-frequency sampling resolution to comprehensively chart the induction, maintenance and recovery of the mammalian circannual interval timer (**Fig. 1H**). Biodare2.0 identified 290 (Dataset S1) transcripts in the mediobasal hypothalamus as endogenously rhythmic. Density heat-maps show an early, and robust upregulation of transcripts by 8 weeks SP associated with the circannual interval induction (**Fig. 1F**). Consistent with previous reports (Petri et al., 2016; Bao et al., 2019; Yoshimura et al., 2003), *Dio3* expression was significantly upregulated and clustered with the initial wave of transcript expression (**Fig. 1F** and *SI Appendix Fig. S1*). A second cluster of transcripts were upregulated during the maintenance phase occurring between 12- and 20-weeks in SP which primarily consisted of somatostatin (*Sst*) and the cognate receptor subtype 5 (*Sstr5*). Increased *Sst* expression coincided with the reduction in body and adipose tissue mass reflecting the conserved role in inhibiting growth and metabolism (*SI Appendix Figs. S1* and *S2*). Finally, a third wave of transcripts became upregulated during the recovery phase, and largely resembled the molecular landscape in LP hamsters (**Fig. 1F**). qPCR assays for *Dio3* and *Sst* expression reflected the sequencing transcript count pattern providing independent replication of the sequencing analyses (**Fig. S1**). In the pituitary gland, 250 transcripts were identified as rhythmic and included genes associated with secretagogue cells such as somatotrophs (*Gh*) and lactotrophs (*Pr1*) (**Fig. 1g** and *SI Appendix Fig S3* and Dataset S2). Similar rhythmic expression was identified in other hypothalamic nuclei including the paraventricular nucleus (518 transcripts) (*SI Appendix Fig. S4* and *S5* and Dataset S3), and the dorsomedial hypothalamus (374 transcripts) (**Fig. S4** and *SI Appendix S6* and Dataset S4).

***Dio3*-dependent changes in triiodothyronine induced body mass loss via *Sst* expression**

Treatment of Djungarian hamster with the somatostatin agonist pasireotide decreases body mass and is sufficient to drive changes similar to SP exposure (Dumbell et al., 2017, 2015). Further, certain SST receptor subtypes appear to be involved in driving seasonal torpor in Djungarian hamsters (Scherbarth et al., 2015). To determine if hypothalamic *Sst* expression reflects the programmed rheostatic change in energy state governed by circannual interval timing or homeostatic cues, tissues were collected from *ad libitum* (AL) fed hamsters under either LP control condition or the maintenance phase (12-week SP). To induce a negative energetic state, hamsters experienced an acute overnight food restriction (FR; 16hrs) (**Fig. 2A**). As anticipated, body mass decreased approximately 30% after exposure to SP (**Fig. 2B**, closed symbols); homeostatic negative energy state challenges by food restriction reduced body mass by 1.5g on average for both LP (open symbols) and SP conditions (**Fig. 2C**). Epididymal adipose tissue mass was higher in LP compared to SP but did not decrease in response to overnight FR (*SI Appendix Fig. S7*).

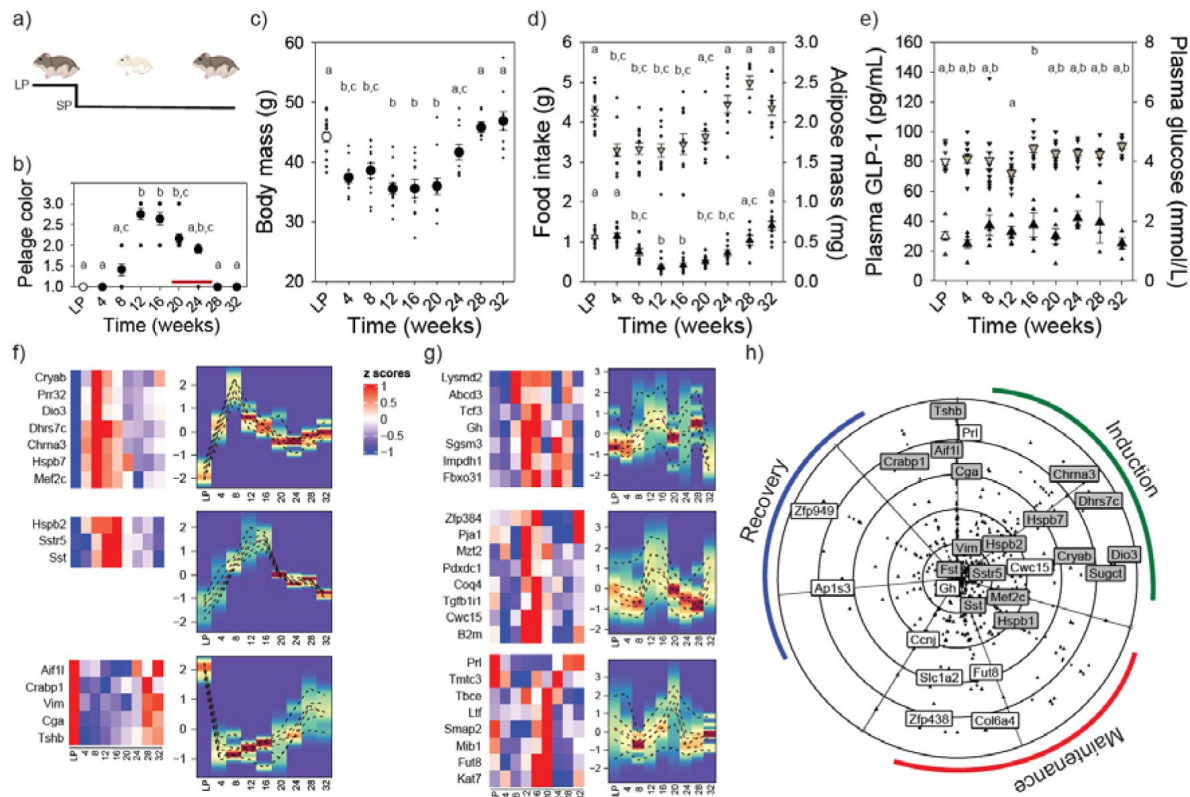


Figure 1.

Molecular basis of circannual interval timing for morphology and physiology.

Experimental design in which Djungarian hamsters were kept in long photoperiod (LP) or transferred to short photoperiod (SP) (a). SP induced pelage color change from LP agouti to winter white after 12-weeks which reversed to agouti by 28-weeks ($F_{8,91} = 50.77$; $P < 0.001$). Created with *BioRender.com* <https://www.biorender.com/>. (b). Torpor was identified in hamster between 12- and 20-weeks SP exposure indicate by red line (b). SP exposure induced significant reduction in body mass ($F_{8,91} = 13.428$; $P < 0.001$) (c), food intake ($F_{8,91} = 10.860$; $P < 0.001$; denoted as downward triangles) (d) and adipose mass ($F_{8,91} = 27.929$; $P < 0.001$; denoted as upward triangles) (d). Plasma GLP-1 did not vary in response to SP manipulation; denoted as upward triangles (e). SP exposure resulted in significant changes in plasma glucose around the onset of torpor; denoted as downward triangles ($F_{8,91} = 3.117$; $P < 0.05$) (e). BioDare2.0 heatmaps of mediobasal hypothalamus (f) and pituitary gland (g) transcripts from Djungarian hamster collected at 4-week SP intervals. Transcripts identified as highly rhythmic ($FDR < 0.1$) showed three distinct phases of expression that coincide with the induction, maintenance and recovery of circannual interval timing. Deiodinase type-3 (*Dio3*) was upregulated during the induction phase, whereas transcripts associated with energy stability (e.g., somatostatin [*Sst* and *Sstr5*]) were upregulated during the maintenance phase. All rhythmic transcripts reverted to the LP condition by 28-weeks SP exposure. Polar scatter chart of significant transcripts from mediobasal hypothalamus and pituitary gland provide a comprehensive seasonal clock for mammalian circannual interval timing across neuroendocrine tissues (h). Green line indicates the induction phase, red line indicates the maintenance phase, and the blue line represents the recovery phase. Letters denote significant differences between treatment groups ($P < 0.05$) (b-e). In panel f and g, the scale bar represents transcript expression as z-scores from 1 (upregulation in red) to -1 (down regulation in blue). Density heatmaps are adjacent to transcript expression heat maps and display the average z-score expression of each individual cluster on the y-axis, and the graph shows percentile lines and density of z-score expression.

Plasma insulin was significantly reduced in response to SP treatment and FR suggesting circulating levels are an output of both rheostatic and homeostatic signals (Fig. S7). The two manipulations permit the ability to dissect the impact of rheostatic changes associated with circannual interval timing from those involved in short-term homeostatic changes in energy balance. Mediobasal hypothalamic *Sst* expression was significantly increased in SP conditions, but insensitive to homeostatic challenges in energy state (Fig. 2D). In contrast, mediobasal hypothalamic *Npy* expression was significantly upregulated in food restricted hamsters but was similar across LP and SP conditions (Fig. 2E). *Prl* expression was higher in LP compared to SP conditions (Fig. 2F). *Gh* expression did not change in response to either SP induced circannual changes in energy state, or homeostatic manipulations (Fig. S7). To establish whether mediobasal hypothalamic *Sst* expression is regulated by upstream *Dio3*-dependent changes in local thyroid hormone catabolism, SP housed hamsters received sub-cutaneous injections with either vehicle or triiodothyronine. A single triiodothyronine injection in SP hamsters was sufficient to reduce *Sst* expression compared to vehicle controls (Fig. 2G). These data indicate SP induced *Dio3* expression removes T3-dependent inhibition of *Sst* expression leading to long-term inhibition of body mass during the maintenance phase of circannual interval timing of rheostatic energy balance.

***Dio3* disfunction reduces the period of circannual interval timing**

Next, we sought to establish the functional role of *Dio3* signaling for circannual interval timing of energy rheostasis in hamsters. Targeted genomic mutations of the *Dio3* gene localized to the MBH were achieved via intracerebroventricular (ICV) injection of CRISPR-Cas9 (*Dio3^{cc}*; Fig. S8) to assess the functional role for circannual interval timing. Control hamsters were administered a blank crispr-cas9 construct (*Dio3^{wt}*). Two weeks following surgery, hamsters were transferred to SP and programmed circannual changes in body mass (Fig. 3B) and pelage were monitored (SI Appendix Fig. S8). *Dio3^{cc}* hamsters had slower body mass loss and regained body mass significantly quicker during the recovery phase, compared to *Dio3^{wt}* controls. Similarly, the change in pelage color occurred later in *Dio3^{cc}* compared *Dio3^{wt}*, and *Dio3^{cc}* was never observed to develop full white pelage (SI Appendix Fig. S8). Lomb-Scargle period analysis of body mass identified that *Dio3^{cc}* hamsters have notably shorter duration (period = 26.959) compared to *Dio3^{wt}* (period = 32) (Figs. 3C and D). There was no significant difference in terminal white adipose tissue mass, or daily food intake. A subpopulation of hamsters was physiologically nonresponsive (NR) to the SP manipulation and maintained the higher LP body mass (Fig. 3E). *Dio3* expression in the mediobasal hypothalamus of these NR hamsters remained exceptionally low compared to responsive SP hamsters at 8- and 12-weeks following exposure to SP (Fig. 3F). These data suggest that the inability to increase *Dio3* expression in NR hamsters acts as a form of naturally occurring disfunction that prevents SP induction of circannual interval timing.

Discussion

The molecular and cellular basis of circannual clocks and timers in vertebrates is not well characterized (Helm and Stevenson, 2014). Here, we sequenced the transcriptomes of three hypothalamic regions and the pituitary gland, key neuroendocrine structures which govern programmed circannual rheostatic and homeostatic processes. Our approach provided a high sampling frequency of the circannual interval timer in Djungarian hamsters. The findings produced a robust and comprehensive neural and molecular database which facilitates the delineation of a circannual interval timer in mammals. Several novel transcripts displayed a close correlation with the induction phase of the photoperiodic response. Among these was *Dio3*, conducive with the removal of thyroid hormone (e.g., thyroxine, triiodothyronine) being a key step in the transition to the winter phenotype. The photoinduced change in *Dio3* expression induces a long-term change in physiology and morphology and our results demonstrate that it is insensitive

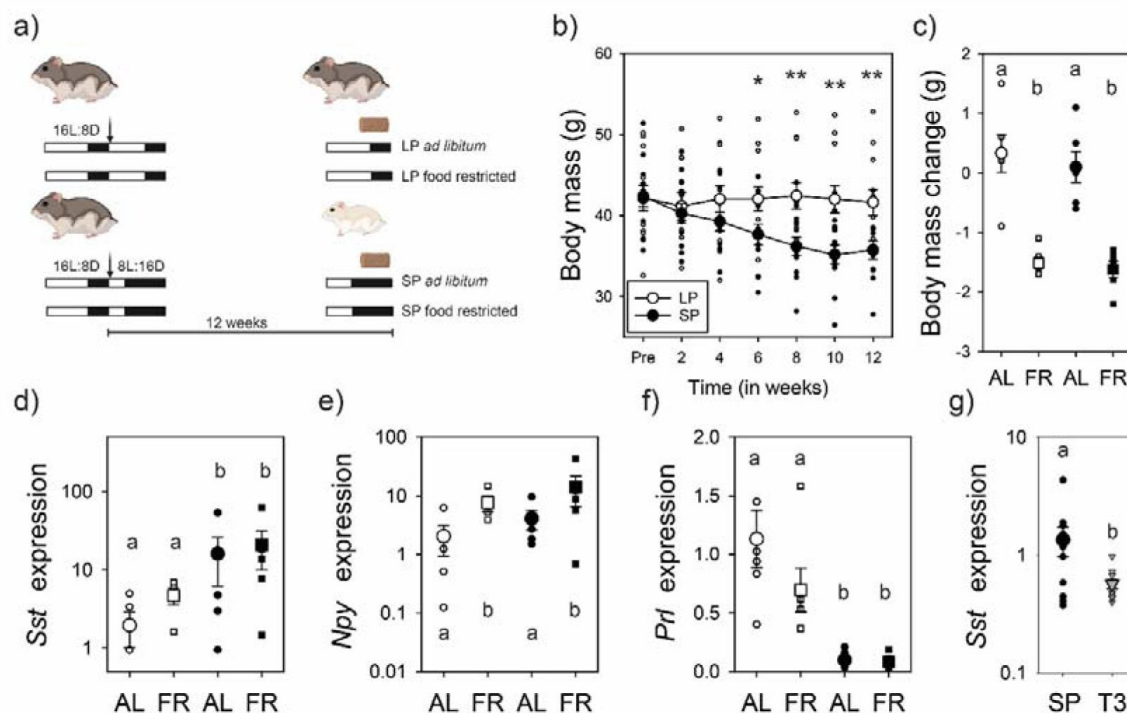


Figure 2.

Somatostatin expression reflects programmed circannual interval timing that is dependent on *Dio3* regulation of local triiodothyronine signaling.

The experimental design in which SP exposure induced rheostatic reduction in body mass after 12-weeks, and then hamsters experienced either an overnight food restriction (FR) for 16 hours or maintained food ad libitum (AL) (a). SP induced a significant reduction in body mass ($F_{1,22} = 3.85$; $P < 0.01$). Created with *BioRender.com* (b). Food restriction further reduced body mass ($F_{1,22} = 25.46$; $P < 0.001$) (c). *Sst* expression in the mediobasal hypothalamus was significantly increased in SP ($F_{1,15} = 5.59$; $P < 0.05$) but did not change after manipulations in nutritional availability (d). *Npy* expression was increased after food restriction ($F_{1,15} = 6.12$; $P < 0.05$) but did not change with SP exposure (e). Prolactin (*Prl*) expression in the pituitary gland was downregulated in response to SP ($F_{1,20} = 28.53$; $P < 0.001$), and insensitive to food restriction. 12-weeks of SP were found to increase *Sst* expression in the hamster mediobasal hypothalamus and levels were significantly reduced in response to a single triiodothyronine (T3) injection (g). * $P < 0.05$ and ** $P < 0.01$ denote significant difference between SP and LP conditions (b). Letters denote significant difference between treatment groups ($P < 0.05$) (c-g).

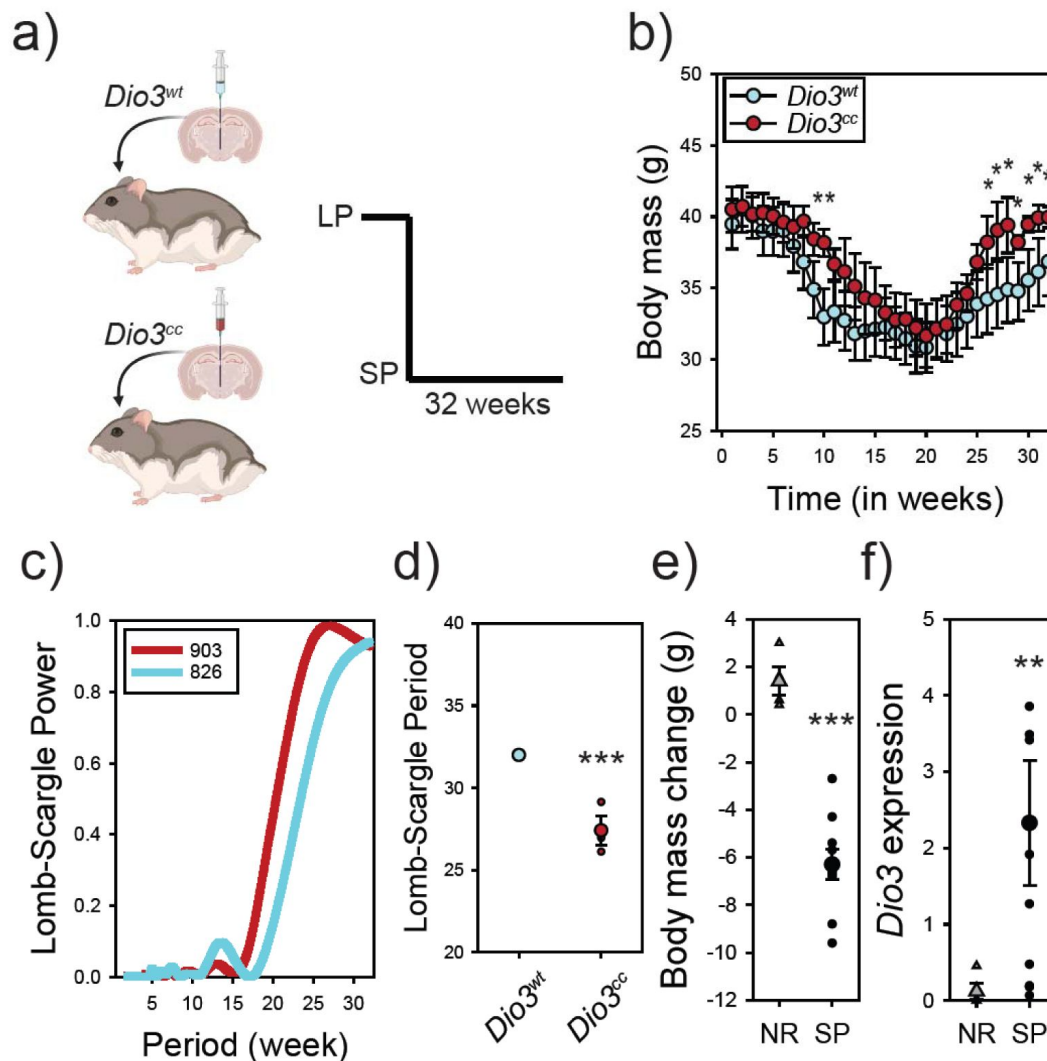


Figure 3.

***Dio3* functions to time circannual interval duration in hamsters.**

Hamsters received intracerebroventricular injections to target *Dio3* expressing cells in the tanycytes localized to the mediobasal hypothalamus (a). Crispr-Cas9 constructs were packaged into lentiviral vectors to generate blank control hamsters (*Dio3*^{wt}) or contain gRNAs that mutated the *Dio3* gene (*Dio3*^{cc}). Hamsters were then exposed to SP conditions and the circannual interval timer was assessed by monitoring body mass. Created with [BioRender.com](https://www.biorender.com) (b). *Dio3*^{cc} hamsters were slower to initiate interval timing as evidenced by higher body mass at 10-weeks SP exposure ($P < 0.001$) and recovered body mass quicker at 26 to 32 ($P < 0.05$) weeks. Lomb-Scargle analyses identified a significant reduction in the period of the circannual interval timer (Welch's t-test; $t_3 = 2.62$; $P < 0.05$) (c,d). Example *Dio3*^{cc} (hamster #903) and *Dio3*^{wt} (hamster #825) are presented in (c). A subpopulation of hamsters did not decrease body mass in response to SP exposure and termed non-responsive (NR) ($t_{12} = 7.12$; $P < 0.001$) (e). *Dio3* expression in the mediobasal hypothalamus of NR hamster was nearly nondetectable after exposure to 8-12 weeks of SP exposure compared to SP control hamsters ($t_{12} = -3.78$; $P < 0.005$) (f). * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

to short-term alterations in response to other environmental cues (e.g., nutritional state). Functional manipulation of *Dio3* expression also delayed the induction and enhanced the reversal of the winter phenotype suggesting that upregulation of *Dio3* acts to establish the period of circannual interval timing.

The transcriptomic data are consistent with evidence that shows *Sst* expression in the arcuate nucleus is one critical event for the maintenance of a reduced rheostatic energy state during the winter season (Marshall et al., 2024 [↗](#); Petri et al., 2016 [↗](#), 2014 [↗](#)). In hamsters, *Sst* expression is upregulated after *Dio3* induction and is associated with the low body mass from week 12-20 before a downregulation in transcript levels coinciding with increased body mass. By increasing circulating triiodothyronine in hamsters maintained in SP conditions, our manipulation indicates that elevated hormone concentrations lead to the reduction in *Sst* expression in the mediobasal hypothalamus (Fig.2 [↗](#)). The overall expression patterns provide the ability to develop a linear series of steps in which LP conditions have high local concentrations of triiodothyronine in the mediobasal hypothalamus (Helfer and Stevenson, 2020 [↗](#); Murphy et al., 2012 [↗](#)) which reduces the levels of *Sst* expression resulting in consistently higher body mass. Exposure to SP induces *Dio3*, which catabolizes triiodothyronine leading to the removal of the inhibition of *Sst* expression and reduction in body mass. Local synthesis of triiodothyronine in the mediobasal hypothalamus by tanycytes is an evolutionary conserved process for timing photoperiod-induced transitions across the animal kingdom (Ebling and Lewis, 2018 [↗](#); Lewis and Ebling, 2017 [↗](#)). Photoperiodic signals derived from the pars tuberalis via thyrotropin-stimulating hormone are essential to regulate *Dio3* (and *Dio2*) expression in tanycytes leading to increased triiodothyronine content in LP conditions for mammals (22,23), birds (Nakao et al., 2008 [↗](#)) and fish (Lorgen et al., 2015 [↗](#)). The current limitation is understanding how SP signals regulate *Dio3* expression. Undoubtedly, the duration of melatonin secretion during the daily nocturnal phase is essential (Helfer et al., 2019 [↗](#)). Future research is necessary to identify the conserved upstream signals that govern the timing of *Dio3* expression.

These data support the existence of a neuroendocrine pathway for the long-term rheostatic regulation, and another for short-term homeostatic control of energy stability (Stevenson, 2024 [↗](#)). The rheostatic pathway consists of triiodothyronine (T3) signaling dependent on a *Dio3* switch to regulate *Sst* expression in the mediobasal hypothalamus whereas short-term regulation of energy stability via nutrient availability is homeostatically regulated by well-established orexigenic (e.g., *Npy*) (Yeo et al., 2021 [↗](#)). In response to the output of both pathways, peripheral effects may be mediated by acute and chronic changes in pancreatic insulin secretion (Fig.4 [↗](#)). Overall, the findings provide a clear neural and cellular circuit for circannual interval timing of energy rheostasis and demonstrate *Dio3* as a gene critical for the control of seasonal life history transitions.

Materials and Methods

Ethics

All procedures were in accordance with the National Centre for the Replacement, Refinement and Reduction of Animals in Research ARRIVE guidelines (<https://www.nc3rs.org.uk/revision-arrive-guidelines> [↗](#)). All procedures were approved by the Animal Welfare and Ethics Review Board at the University of Glasgow and conducted under the Home Office Project License PP5701950.

Animals

All animals were obtained from a colony of Djungarian hamsters (3-8 months of age) kept at the University of Glasgow, Veterinary Research Facility. Animals were held at a room temperature of 21°C, and a humidity of 50%. The colony room was kept on a long photoperiod (16L:8D). All hamsters had food (SDS [BK001]) and water *ad libitum*.

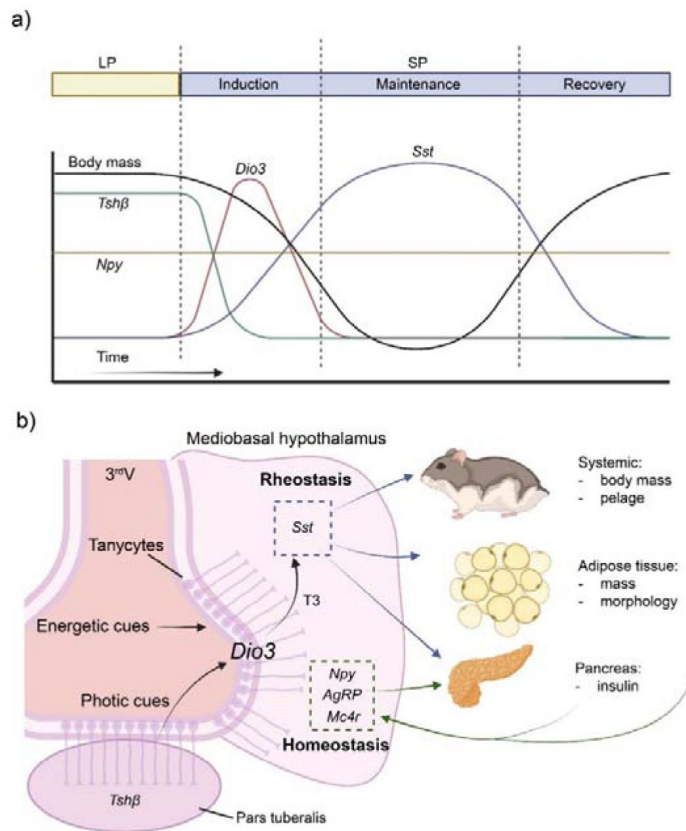


Fig 4.

Schematic model for the mechanisms of circannual interval timing.

(a) Hamsters held in long photoperiod (LP) have high body mass, thyrotropin-stimulating hormone beta (*Tshβ*) and low somatostatin (*Sst*) and deiodinase type-3 (*Dio3*). Transfer to short photoperiod (SP) results in the induction of circannual interval timing characterized by a reduction in body mass, low *Tshβ* in the pars tuberalis and increased *Dio3* expression in tanycytes. The maintenance phase is associated with increased *Sst* expression which serves to inhibit growth. The onset of the recovery phase is associated with a complete reversal in whole organismal physiology to the LP phenotype. (b) tanycytes along the third ventricle in the mediobasal hypothalamus integrate photic cues derived from *Tshβ* and are sensitive to nutritional cues. The triiodothyronine (T3) output signal from tanycytes regulates *Sst* expression leading to long-term programmed rheostatic changes in body mass that serve to control circannual interval timing in multiple physiological systems. Conversely, homeostatic stability is maintained despite large scale seasonal rhythms in body mass. Homeostatic energy balance is established by well characterized circuits that include neuropeptide Y (*Npy*), agouti-related peptide (*AgRP*) and the melanocortin receptor 4 (*Mc4r*). Created with [BioRender.com](https://www.biorender.com)

Experiment 1 – Defining the circannual interval timer in hamsters

Adult male hamsters (N = 100) were required to assess the physiological and molecular basis of circannual interval timing in hamsters. A photoperiod reference group was kept on LP (n=14) (8D:16L) conditions for the duration of the study. The SP treatment groups consisted of hamsters moved to SP (16D:8L) at 4 weeks intervals. The groups consisted of SP for 4 (n=12), 8 (n=12), 12 (n=12), 16 (n=11), 20 (n=12), 24 (n=10), 28 (n=7) and 32 (n=10) weeks. This photoperiod schedule reliably captures the induction, maintenance, and recovery of circannual interval timing in hamsters (Prendergast et al., 2002 [\[1\]](#)). Pelage score was determined using visual assessment and defined scale in which a score of 1 is the summer dark agouti coat; a score of 2 is a transition between agouti and white; and a score of 3 is a full winter white pelage (Marshall et al., 2024 [\[2\]](#)). Body mass was monitored to the nearest 0.1g using an Ohaus portable scale (TA301). Food intake was measured by weighing the food pellets at the mid-point on two consecutive days to obtain a daily value. Food weight was measured using an Ohaus scale (TA301). Hamsters were killed between 2-4hr after lights on by cervical dislocation followed by exsanguination. Adipose tissue was immediately dissected and measured using a Sartorius cp64 anatomical balance. A terminal blood sample was collected to measure glucose (Accu-check Performa nano blood glucose meter). Circulating levels of insulin and total glucagon-like peptide 1 (GLP1) using ELISA (MesoScale Discovery, UK) at Core Biochemical Assays laboratories, Cambridge, UK. Brain and pituitary gland tissue was quickly extracted and frozen on powdered dry ice and stored at -70°C. Epididymal white adipose tissue was dissected and weighed using Sartorius cp64 anatomical balance and stored at -70°C.

Brains were cut into 200µm coronal sections using a Leica CM1520 cryostat. Anatomical structures (optic tract to the infundibular stem; Bregma -2.12mm to -3.80mm) were used to isolate the mediobasal hypothalamus (MBH), dorsomedial hypothalamus (DMH) and the paraventricular nucleus (PVN). Bilateral tissue punches were performed using an integra Miltex 1mm disposable biopsy punch. Tissue punches were stored at -70°C until transcriptomic sequencing and confirmatory qPCR analyses (see below).

Experiment 2 – Dissociating mechanisms that govern circannual rheostatic versus homeostatic energy stability

A total of N=35 adult male hamsters were used to determine molecular signatures of rheostatic versus homeostatic energy stability. A LP control group was kept on long photoperiod for the duration of the study (n = 18). A subset of hamsters was moved to SP (n = 17) for 12 weeks to obtain animals in the maintenance phase of circannual interval timing. Food (SDS [BK001]) and tap water were provided *ad libitum* to both LP and SP animals for the 12 weeks duration. Body mass was measured every two weeks using a Traveler Ohaus portable balance. On the final night of the experiment, 50% of hamsters (i.e., LP = 9, SP = 9) were kept on food and water *ad libitum*. The other 50% of hamsters (i.e., LP = 9, SS = 8) served as the homeostatic treatment group and received an acute food restriction by removing all food. Overnight food restriction induces a robust negative energy state that reliably results in a 1–2-gram loss in hamster body mass (Bao et al., 2019 [\[3\]](#)). Body mass was measured prior to food restriction, and again before tissue collection on the subsequent day. Hamsters were killed between 2-4hr after lights on by cervical dislocation and then exsanguination. A terminal blood sample was collected to measure insulin (MesoScale Discovery, Core Biochemical Assays laboratories, Cambridge, UK). Body mass and epididymal white adipose tissue mass were measured using a Sartorius cp64 anatomical balance. Brains and pituitary glands were dissected and were immediately placed on dry ice and stored at -70°C. Brains were cut into 200µm sections using a Leica CM1520 cryostat. Anatomical structures (optic tract to the infundibular stem; Bregma -2.12mm to -3.80mm) were used to isolate the MBH. Bilateral tissue punches were performed using an integra Miltex 1mm disposable biopsy punch. Brain tissue punches and pituitary gland samples were stored at -70°C until qPCR assay (see below).

Experiment 3 – Sufficiency of triiodothyronine to induce LP neuroendocrine state

Adult male hamsters (n = 35) were used to determine the impact of triiodothyronine on *Sst* expression. Hamsters were transferred from LP to SP for 12 weeks (n = 24) to initiate circannual interval timing and establish the maintenance phase. Hamsters were divided into two groups for the last week of SP treatment. A SP reference group received subcutaneous injections of 0.9% w/v saline for 7 days (n = 10). To determine the sufficiency of triiodothyronine to regulate *Sst* expression, hamsters (n=11) were administered a single subcutaneous injection of 5µg/100ul triiodothyronine (Merck 102467157) prior to lights off on the final night. Subcutaneous injections of triiodothyronine are well established to regulate transcript expression in the hamster hypothalamus (Banks et al., 2016 [↗](#); Bao et al., 2019 [↗](#); Stevenson et al., 2014 [↗](#)). The subsequent day hamsters were sacrificed by cervical dislocation followed by exsanguination between 4-5hr after lights on. Terminal body mass and epididymal adipose tissue mass were recorded using a Sartorius cp64 anatomical balance. Brains were dissected and stored immediately at -70°C until sectioning. Brains were cut into 200µm sections using a Leica CM1520 cryostat. Anatomical structures (optic tract to the infundibular stem; Bregma -2.12mm to -3.80mm) were used to isolate the MBH. Bilateral tissue punches were performed using an integra Miltex 1mm disposable biopsy punch. Tissue punches were stored at -70°C until qPCR assays (see below).

Experiment 4 – Functional manipulation of *Dio3* expression in hamsters

Targeted neuroanatomical injections, using highly precise Guide RNA (gRNA) against *Dio3* and Crispr-Cas9 vectors was essential to ensure the long-term inhibition of *Dio3* function in the MBH in hamsters; an approach necessary to reliably examine the functional significance of *Dio3* for the duration of circannual interval timing. Custom-made Crispr-cas9 constructs were packaged in a lentiviral vector by Merck Life science UK. Lentiviral vectors are established to effectively transfect in adult Syrian hamsters (Gao M et al., 2014 [↗](#)) and in Siberian hamsters (Munley K et al., 2022 [↗](#)). The lentiviral vector used was U6-gRNA:ef1a-puro-2A-Cas9-2A-tGFP (Sigma All-in-One vector). gRNA for *Dio3* were designed using CHOPCHOP (Labun et al., 2019 [↗](#)). Three gRNA were designed to target the *Mus musculus Dio3* gene (gRNA1: CGACAACCGTCTGTGCACCCTGG; gRNA2: GTTCCCGCGCTTCTAGGCACGG, and gRNA3: GACCCAGCCGTCGGATGGGTGGG). Only gRNA 1 and 2 were identified to align with 5' end of the Siberian hamster *Dio3* gene and were predicted to align with chromosome 12 at locations 110279543 and 110279375 respectively (Fig. S8 [↗](#)). To check specificity of gRNA sequences, we conducted BLAST using the NCBI dataset. Both gRNA1 and gRNA2 were found to align 100% with *Dio3* and did not align 23/23 with any other region.

Adult male hamsters (n = 10) were selected to assess the impact of *Dio3* genome modification on circannual interval timing. Hamsters were kept on LP conditions to maintain the summer phenotype. Hamsters were then divided into two treatment groups. The control group included hamsters (n=6) that received intracerebroventricular (ICV) injections of blank Crispr-Cas9 constructs that were packaged into a lentivirus (*Dio3*^{wt}; U6-gRNA:ef1a-puro-2A-Cas9-2A-tGFP (Sigma All-in-One vector). The treatment group consisted of hamsters (n=4) that received an ICV injection of Crispr-cas9 constructs that harbored both gRNA1 and gRNA2 (*Dio3*^{cc}).

Intracerebroventricular injections provide the ability to induce neuroanatomically localized manipulations in *Dio3* gene function in the adult brain. Crispr-Cas9 constructs were delivered via stereotaxic ICV injection into the third ventricle to target the MBH tanycytes. Under general anesthesia (5% induction, 2% maintenance of isoflurane mixed with oxygen) a SGE series II syringe was positioned along the midline at Bregma -1.5mm, to a depth of -5.7mm. The anatomical coordinates were refined based on estimates taken from adult hamster brains (Steward et al., 2003 [↗](#)) and tested using fresh hamster cadavers. Analgesia was administered subcutaneously

(5mg/kg Rimadyl and 0.1mg/kg buprenorphine). A small ~1cm incision was made by scalpel to expose the skull and a small hole was drilled using a Bone Micro Drill for Brain surgery (Harvard Apparatus). Lentivirus was administered using a Pump 11 Elite Programmable Syringe injection system (Harvard Apparatus) and 1ul of viral vector (2.5×10^{13} vg/ml) was delivered over 10 minutes at a rate of 0.1ul per minute. To allow viral diffusion and prevent backflow the syringe was kept in place after injection for 1-minute and then the syringe was raised by 1mm and kept in place for another 1-minute.

Animals were then moved to heated home cages and given mashed food to aid recovery. Hamsters were provided with 2-weeks recovery and were monitored to ensure stable body mass, locomotor activity and continued food and water intake.

To assess the impact of *Dio3^{cc}* on circannual interval timing hamsters were moved to SP for 32 weeks and body mass and pelage were measured biweekly. Pelage score was determined using visual assessment and defined scale in which a score of 1 is the summer dark agouti coat; a score of 2 is a transition between agouti and white; and a score of 3 is a full winter white pelage (Marshall et al., 2024 [DOI](#)). Body mass was monitored to the nearest 0.1g using an Ohaus portable scale (TA301). After 32 weeks, hamsters were killed 4-5hr after lights on by cervical dislocation followed by exsanguination. Brains were dissected and stored immediately at -70°C. Brains were cut into 200µm sections using a Leica CM1520 cryostat. Anatomical structures (optic tract to the infundibular stem; Bregma -2.12mm to -3.80mm) were used to isolate the MBH. Bilateral tissue punches were performed using an Integra Miltex 1mm disposable biopsy punch. Tissue punches were stored at -70°C. DNA was extracted from dissected tissues using the DNeasy Blood & Tissue Kit (QIAGEN) as per manufacturer's instructions. The *Dio3* gene was amplified using OneTaq Quick-Load 2X mastermix, as per manufacturer's instructions (New England Biolabs). Amplification was achieved using a SimpliAmp thermal cycler (Applied Biosystems) the following thermal cycling conditions (94°C for 30 seconds, 94°C for 15 seconds, 62°C for 30 seconds, 68°C for 3 minutes) for 45 cycles, 68°C for 5 minutes, 4°C until further analysis. The resultant amplicon was purified using the Qiaquick PCR purification kit (Qiagen) as per manufacturer's instructions. Isolated *Dio3* PCR amplicons were sequenced using the Eurofins LightRun sanger sequencing service. The forward primer provided to Eurofins was CATGCTCCGCTCCCTGCTGCTTCA (5'-3'). CRISPR modified transcripts were aligned to a reference wild type transcript (animal #821) using the Tracy sanger basecaller and aligner (Rausch et al., 2020 [DOI](#)). Alignments were visualized using the Ugene integrated bioinformatics tool (Okonechnikov et al., 2012 [DOI](#)). In case mutation was not obvious, an additional analysis was carried out using the associated online resource SAGE (<https://www.gear-genomics.com/> [DOI](#)), to investigate the decomposition error of the modified transcripts with the wild-type control (821). Four *Dio3^{wt}* hamsters had *Dio3* sequences that fully matched the reference genome (Fig. S8c [DOI](#)). Two hamster *Dio3* sequences did not automatically align, and the raw reads were visually inspected. No mutations or sequence mutations were detected in the raw sequence trace. Three *Dio3^{cc}* treated hamsters had *Dio3* sequences with evidence of genomic mutation evidenced by low decomposition error (SI Appendix Figs. S8C and D). One *Dio3^{cc}* treated hamster did not show any genomic mutation (905) and was removed from the final analysis. Hamster 905 had a body mass circannual interval period of 32 weeks and we propose the animal represents a false positive control.

Experiment 5 – Naturally occurring variation in *Dio3* expression and photoperiodic response

A subset of hamsters does not physiologically respond to SP treatment and are classified as photoperiodically non-responsive (NR) (Przybylska-Piech and Jefimow, 2022 [DOI](#)). Adult hamsters (n = 14) were held in LP or SP for 8 weeks. A group of male non responders (n = 4) were identified due to the absence of body mass and pelage change in response to SP (Fig. 3e [DOI](#)). A reference

group of male SP responders ($n = 10$) were collected. MBH was extracted, RNA isolated and cDNA synthesized. qPCR was carried out for *Dio3* and *Sst* expression. Hamsters were killed between 2-4hr after lights on.

RNA extraction

RNA was extracted using the Qiagen RNEasy plus mini kit. Tissues were homogenized using a Polytron PT 1200 E. RNA was then extracted from homogenized tissue following the manufacturer's instructions. RNA was tested for quality and quantity using an ND-1000 Nanodrop.

Next Generation Sequencing and data processing

Oxford Nanopore Sequencing was used to carry out transcriptomic sequencing. RNA from MBH, DMH and PVN samples was sequenced using the PCR-cDNA barcoding kit following manufacturer's instructions (SQK-PCB109; Oxford Nanopore). RNA from the pituitary gland samples was synthesized using the direct cDNA sequencing kit following manufacturer's instructions (SQK-DCS109 with EXP-NBD104; Oxford Nanopore). Bioinformatic analysis was carried out on Linux within a conda environment. Raw reads (Fast5) were demultiplexed and basecalled using guppy basecaller (4.2.1). Porechop (0.2.4) was used to remove adapters from reads and Filtlong (v0.2.0) was used to filter for quality and read length (length: >25 base pairs, quality score > 9). Transcripts were aligned to *Mus musculus* genome (GRCm39) using Minimap2 (Li, 2018 [\[link\]](#)). Previously, rodent genomes have been demonstrated to show significant similarity, such that cross-species transcriptomic analysis appears feasible (Przybylska-Piech and Jefimow, 2022 [\[link\]](#)). Transcript expression levels were generated using Salmon (v0.14.2) (Patro et al., 2017 [\[link\]](#)) and EdgeR (v3.24.3) (Robinson et al., 2010 [\[link\]](#)) was used to filter lowly expressed transcripts.

cDNA synthesis and qPCR

RNA was transformed into cDNA for qPCR analysis using superscript III (Invitrogen) as per manufacturer's instructions. Quantification of transformed cDNA was achieved using Brilliant II SYBR Mastermix (Agilent). Stock forward and reverse primers (100pmol/μl) were mixed and diluted in nuclease free water to 20pmol/μl. A working SYBR mixture was prepared by mixing 1-part primer mixture per 24-part SYBR mastermix. Reaction mixtures were prepared in wells on a 96-well plate by mixing 4.8μl of normalised sample and 4.8μl of SYBR working mix. All reactions were performed in duplicate. Primer and qPCR parameters are outlined in **Table S1** [\[link\]](#). qPCR reactions were carried out in a Stratagene Mx3000P thermal cycler. Cycling conditions utilized were in sequence; at 95°C for 5 minutes (denaturing), 40 cycles at (95°C for 30 seconds, X°C for 1 minute – see table, 72°C for 30 seconds with fluorescent measurement at end), 95°C for 1 minute, 55°C for 30 seconds increasing to 95°C (Melt curve analysis). Melt cure analysis was used to determine specificity of amplification. Data was analyzed for meanCT, efficiency and variability using PCR Miner (Zhao and Fernald, 2005 [\[link\]](#)). Logfold CT was calculated using the $\Delta\Delta CT$ method using 18s and/or *Hrpt* reference transcripts as these transcripts are stable across photoperiodic conditions (Stewart et al., 2022 [\[link\]](#)).

Statistical Analysis

Raw data is provided in Data file 1 and sequencing data is provided in **Table S1** [\[link\]](#)-S4. Physiological and qPCR data sets were tested for normality using the Shapiro test and transformed by log transformation if normality was violated. Statistical significance was determined using a two-way ANOVA. Post hoc analysis of significant data was achieved by t-test. Type 1 error was minimized by utilizing the Bonferroni adjustment on post-hoc analyses. Data were analyzed using non-linear regression for rhythmicity using the online resource BioDare 2.0 (Zielinski et al., 2014 [\[link\]](#)) (biodare2.ed.ac.uk). The empirical JTK_CYCLE method was used for detection of rhythmicity and the classic BD2 waveform set was used for comparison testing. Rhythmicity was determined by a Benjamini-Hochberg controlled false discovery rate (BH corrected FDR < 0.1). The false discovery rate of this data set was controlled by applying the Benjamini-Hochberg method (Benjamini and

Hochberg, 1995 [\[1\]](#)). Clusters within the data set were identified using the gap-statistic and clustered using k-means clustering. Gene ontology analysis was carried out using ShinyGO v0.77 (Ge et al., 2020 [\[2\]](#)).

Supplementary Figures

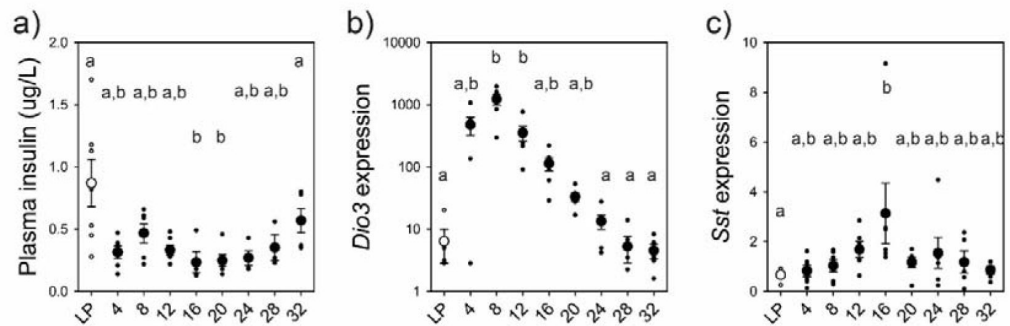


Fig. S1.

Plasma insulin reflects changes in transcript expression.

(A) Plasma insulin of Siberian hamsters across 32-weeks of short photoperiod exposure ($H_8 = 20.401$, $P = 0.009$). (B) Expression of *Dio3* from the MBH of Siberian hamsters, assessed by qPCR, across 32-weeks of short photoperiod exposure ($H_8 = 39.771$, $P < 0.001$). (C) Expression of *Sst* from the MBH of Siberian hamsters, assessed by qPCR, across 32-weeks of short photoperiod exposure ($H_8 = 16.002$, $P = 0.042$). Letters denote significance between groups.

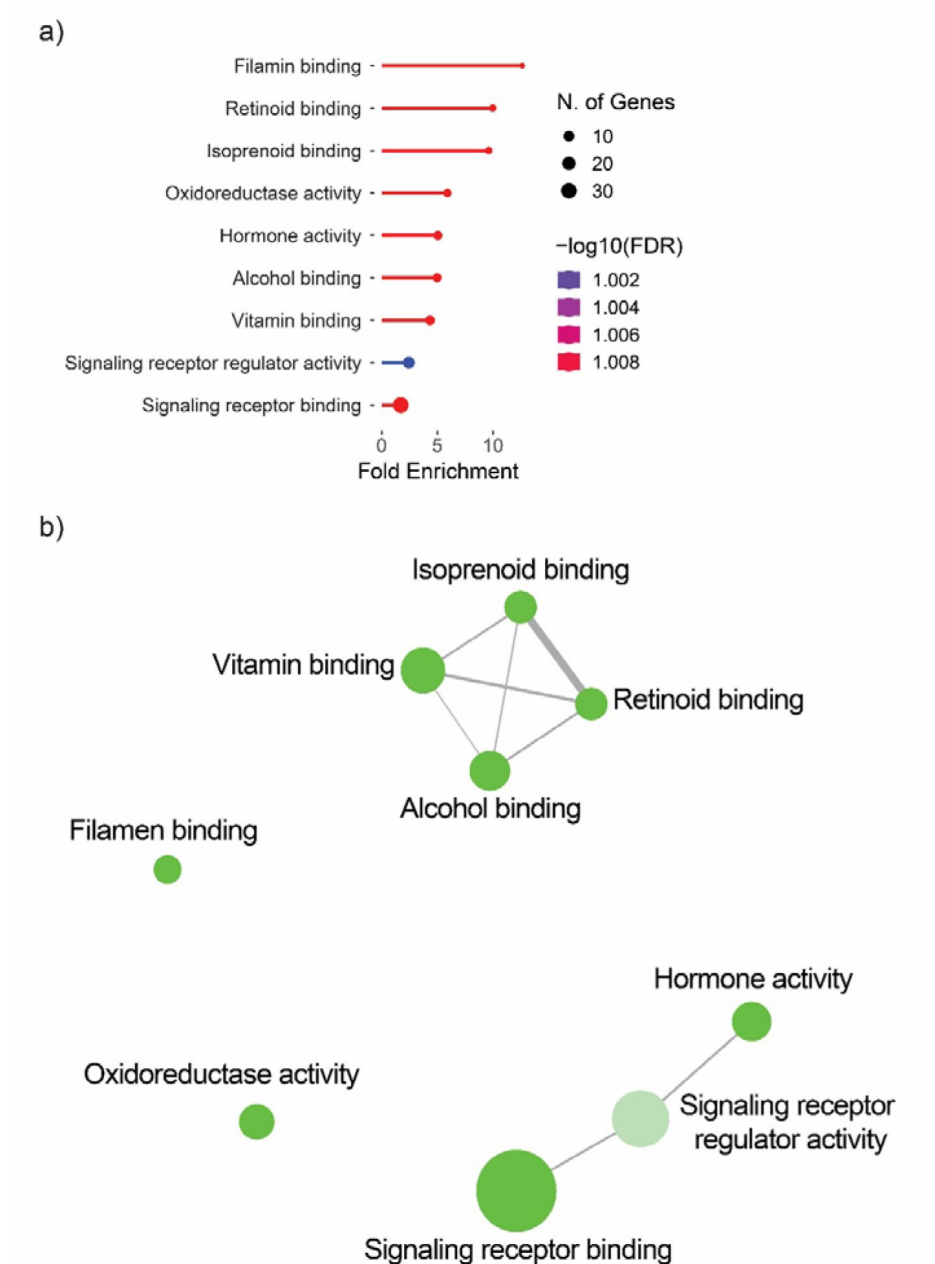


Fig. S2.

Gene ontology analysis of MBH sequencing reveals well known seasonal pathways.

(A) Gene ontology enrichment analysis of significant transcripts from sequencing of Djungarian hamster MBH across 32-weeks of short photoperiod exposure. (B) Interaction network of enriched gene ontology terms. Size of individual points represent the number of transcripts in each term. Thickness of interactions represents shared transcripts between terms.

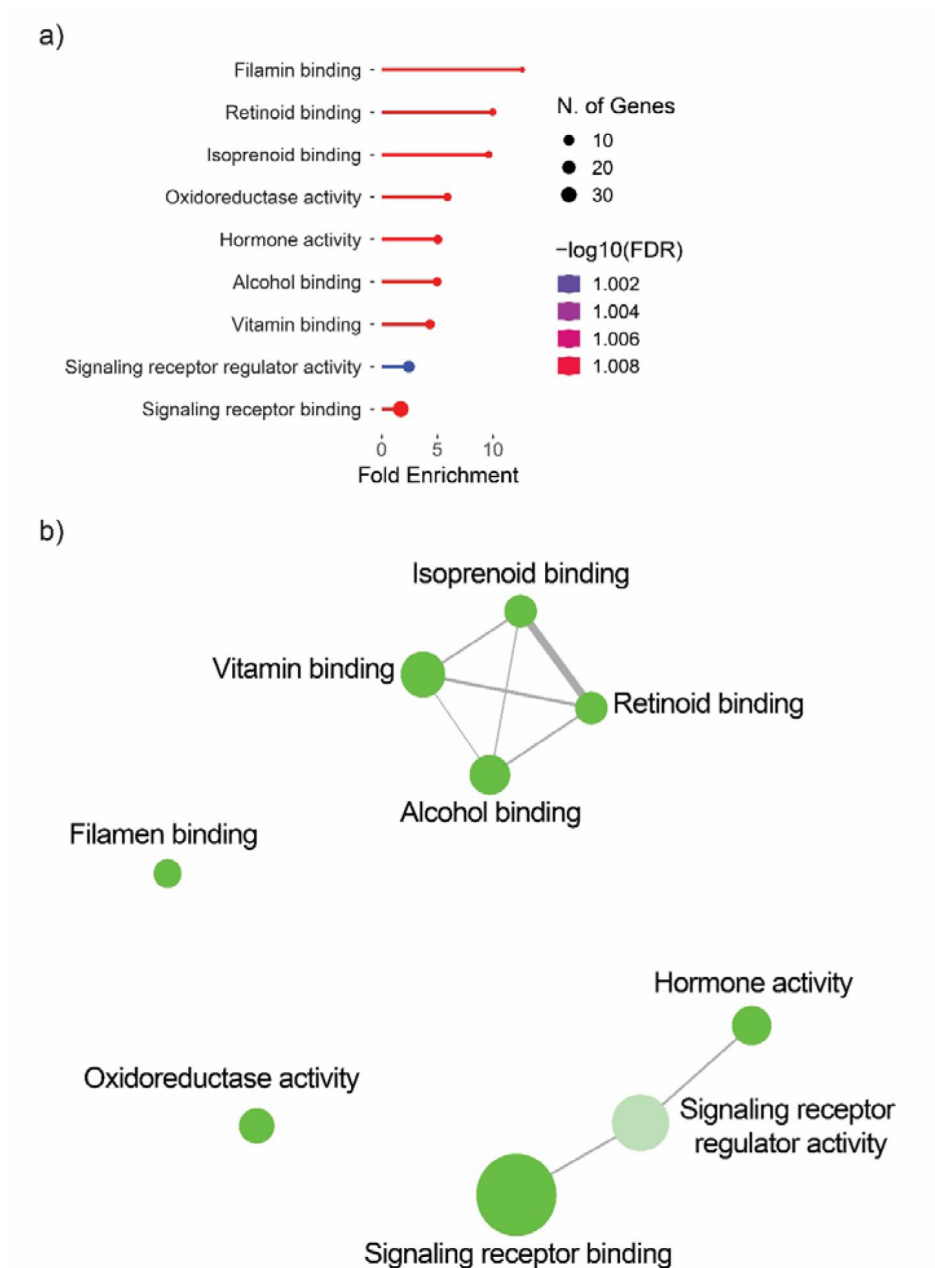


Fig. S3.

Gene ontology analysis of pituitary gland unveils potential mechanisms for seasonal changes in protein processing and release.

(A) Gene ontology enrichment analysis of significant transcripts from sequencing of Djungarian hamster pituitary gland across 32-weeks of short photoperiod exposure. (B) Interaction network of enriched gene ontology terms. Size of individual points represents the number of transcripts in each term. Thickness of interactions represents shared transcripts between terms.

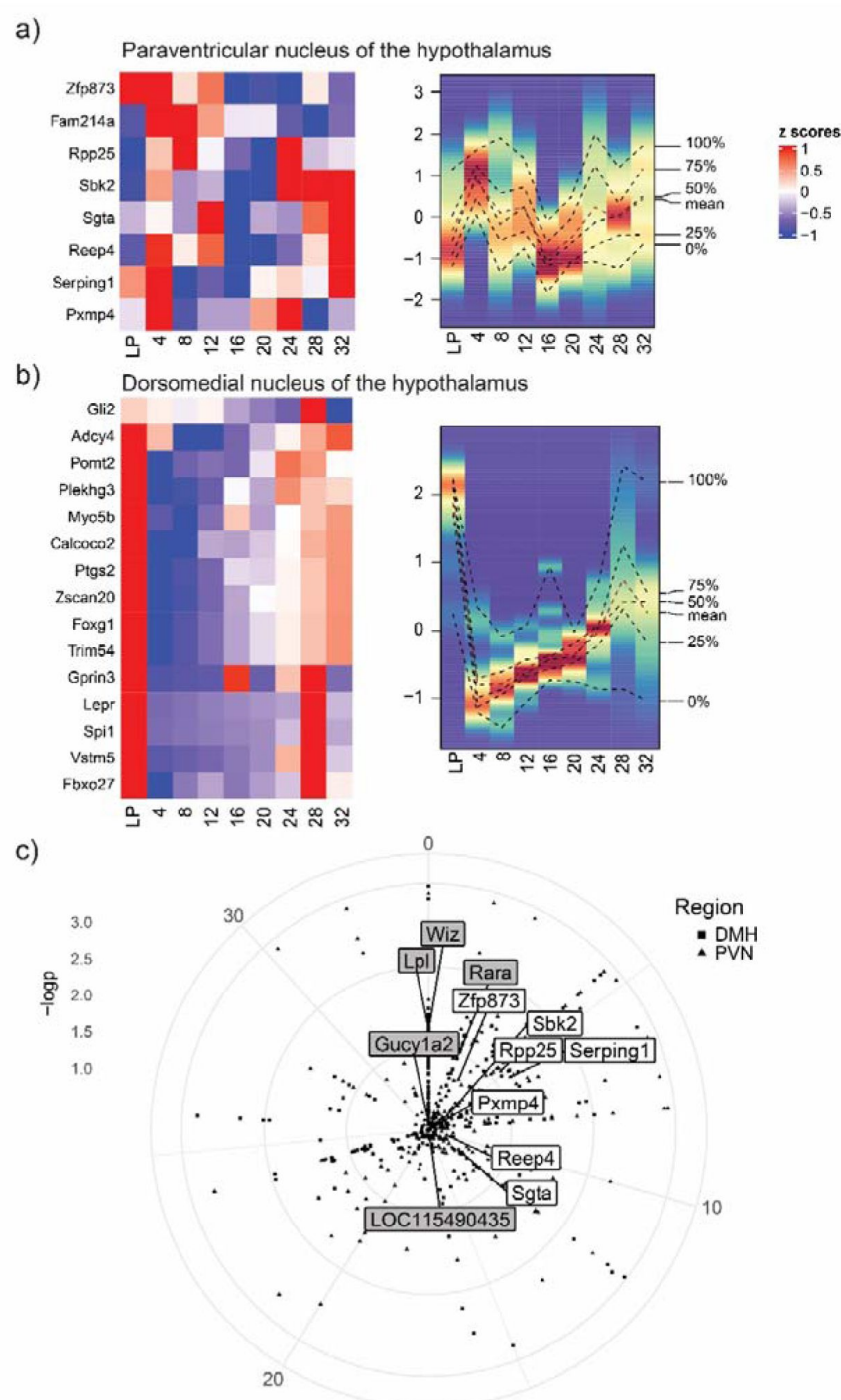


Fig. S4.

Paraventricular and dorsomedial hypothalamus sequencing unveils novel transcripts and widespread seasonal interval timing within the hypothalamus.

(A-B) Heatmaps of significant ($FDR < 0.1$) Djungarian hamster transcripts displaying sine or cosine rhythmicity were selected from (A) paraventricular hypothalamus and (B) dorsomedial hypothalamus. (C) Polar scatter chart of significant transcripts from Djungarian hamster paraventricular and dorsomedial hypothalamus, displaying peak of expression and $-\log(FDR)$.

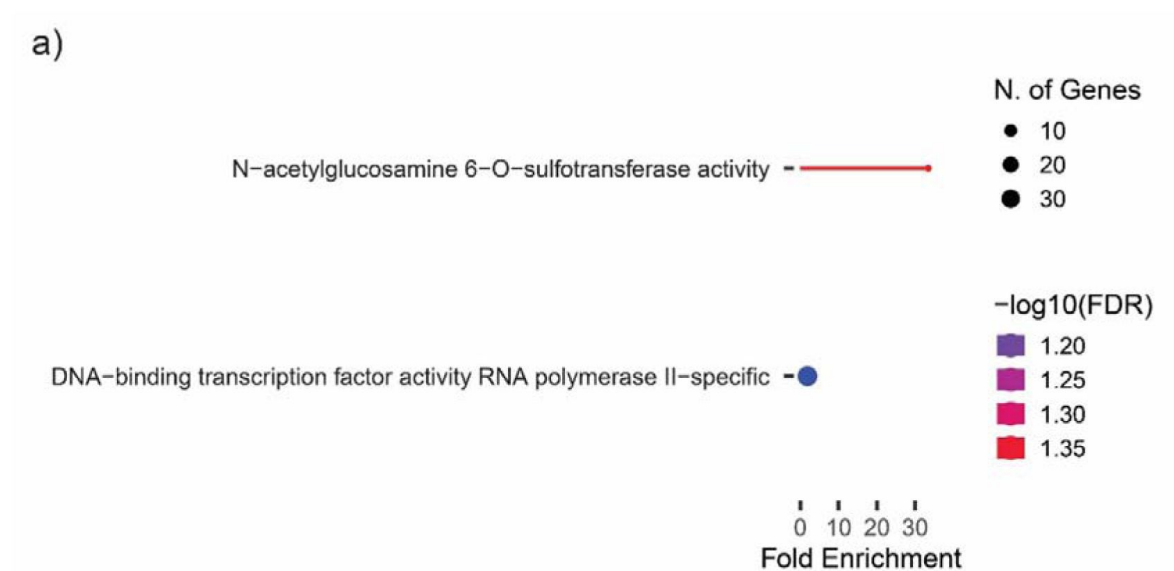


Fig. S5.

Gene ontology analysis of paraventricular hypothalamic sequencing.

(A) Gene ontology enrichment analysis of significant transcripts from sequencing of Djungarian hamster paraventricular hypothalamus across 32-weeks of short photoperiod exposure.

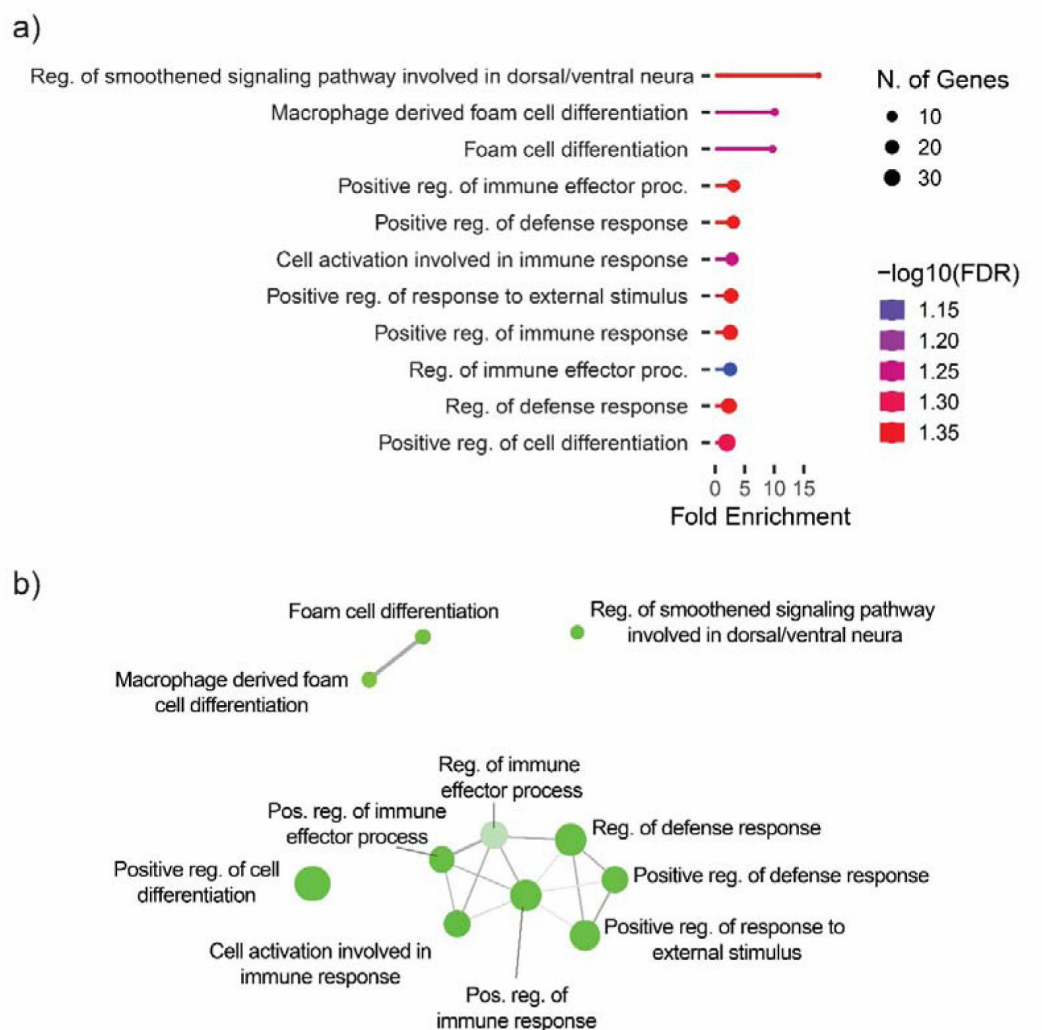


Fig. S6.

Gene ontology analysis of dorsomedial hypothalamus suggests widespread immune involvement and cellular differentiation.

(A) Gene ontology enrichment analysis of significant transcripts from sequencing of Djungarian hamster dorsomedial hypothalamus across 32-weeks of short photoperiod exposure. (B) Interaction network of enriched gene ontology terms. Size of individual points represents the number of transcripts in each term. Thickness of interactions represents shared transcripts between terms.

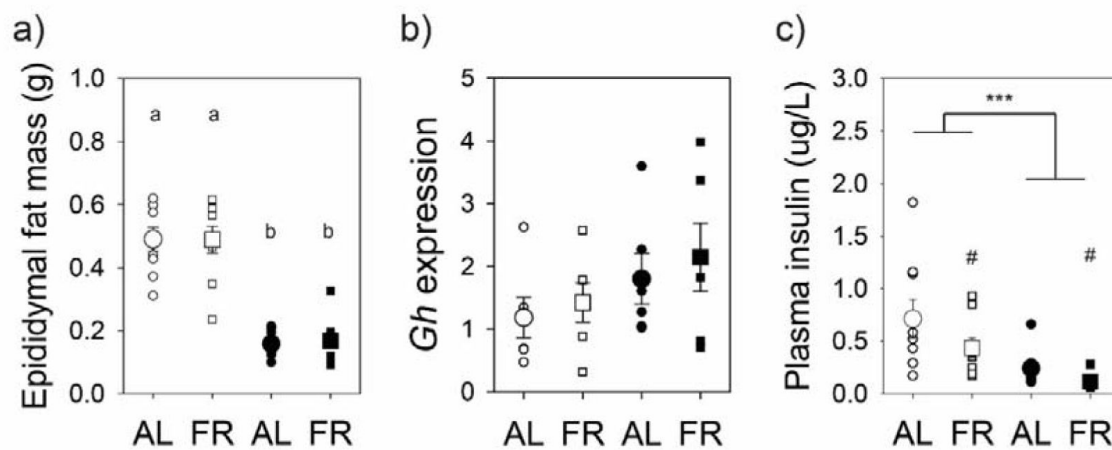


Fig. S7.

Rheostatic mechanism controlling body mass change in Djungarian hamsters.

(A) Epididymal fat mass in long- and short-photoperiod exposed Djungarian hamsters ($F_{1,20} = 59.136$; $P < 0.001$) after an overnight food restriction, or ad libitum, feeding protocol ($F_{1,22} = 2.738$; $P = 0.114$). (B) Pituitary gland expression of *Gh* from in long- and short-photoperiod exposed Djungarian hamsters ($F_{1,20} = 2.738$; $P = 0.114$) after an overnight food restriction, or ad libitum, feeding protocol, assessed by qPCR ($F_{1,20} = 0.518$; $P = 0.480$). (C) Plasma insulin in long- and short-photoperiod exposed Djungarian hamsters ($F_{1,20} = 29.865$; $P < 0.001$) after an overnight food restriction, or ad libitum, feeding protocol ($F_{1,22} = 6.863$; $P < 0.05$). Letters denote significant main effect of photoperiod (a). Asterisk indicates significant effect of photoperiod ($P < 0.005$), and hashtag denotes significant main effect of food restriction ($P < 0.05$).

Gene	Primer	Size	Temp	Melt
Reference	-	-	-	-
<i>18s</i>	GCTCCTCTCCTACTTGGATAACTGTG CGGGTTGGTTTTGATCTGATAAATGCA	111	62°C	80°C
<i>Hrpt</i>	AGTCCCAGCGTCGTGATTAGTGATG CGAGCAAGTCTTTCAGTCCTGTCCA	141	62°C	76°C
Mediobasal hypothalamus				
<i>Npy</i>	CCAGGCAGAGATACGGCAAGAGATC CCATCACCACATGGAAGGGTCC	119	60°C	81°C
<i>Sst</i>	GAAGTCTCTGGCGGCTGCTG CAGCCTCATTTTCATCCTGCTCCG	145	60°C	85°C
<i>Dio3</i>	CATGCTCCGCTCCCTGCTGCTTCA CAGGGTGACACAGACGGTTGTC	251	62°C	85°C
Pituitary Gland				
<i>Prl</i>	TCCGGAAGTCCTTCTGAACC CGCAGGCAGCGAATCTTATTG	300	60°C	82°C
<i>Gh</i>	ACCTACAAAGAGTTTGAGCGTG ATGAGCAGCAGCGAGAATCG	167	58°C	85°C

Table S1.

Djungarian hamster qPCR primers sequences and thermal cycling profiles.

Data availability

All data are available in Dataset S1-S5. The code is available via github repository (https://github.com/CS-Gla/Hamster_seasonal_sci_cs_2024). Sequencing information was deposited in GEO (GSE274003).

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Additional files

Dataset S1

Dataset S2

Dataset S3

Dataset S4

Dataset S5

Additional information

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Reviewer #1 (Public review):

Circannual timing is a phylogenetically widespread phenomenon in long-lived organisms and is central to the seasonal regulation of reproduction, hibernation, migration, fur color changes, body weight, and fat deposition in response to photoperiodic changes. Photoperiodic control of thyroid hormone T3 levels in the hypothalamus dictates this timing. However, the mechanisms that regulate these changes are not fully understood. The study by Stewart et al. reports that hypothalamic iodothyronine deiodinase 3 (Dio3), the major inactivator of the biologically active thyroid hormone T3, plays a critical role in circannual timing in the Djungarian hamster. Overall, the study yields important results for the field and is well-conducted, with the exception of the CRISPR/Cas9 manipulation.

Comments on revisions:

The authors have satisfactorily addressed all my comments. I no longer have concerns about the CRISPR/Cas9 experiments which have been conducted properly and are now reported appropriately.

<https://doi.org/10.7554/eLife.106383.2.sa2>

Reviewer #2 (Public review):

Summary:

Several animals and plants adjust their physiology and behavior to seasons. These changes are timed to precede the seasonal transitions, maximizing chances of survival and reproduction. The molecular mechanisms used for this process are still unclear. Studies in mammals and birds have shown that the expression of deiodinase type-1, 2, and 3 (Dio1, 2, 3) in the hypothalamus spikes right before the transition to winter phenotypes. Yet, whether this change is required or an unrelated product of the seasonal changes has not been shown, particularly because of the genetic intractability of the animal models used to study seasonality. Here, the authors show for the first time a direct link between Dio3 expression and the modulation of circannual rhythms.

The work is concise and presents the data in a clear manner. The data is, for the most part, solid and supports the author's main claims. The use of CRISPR is a clear advancement in the

field. This is, to my knowledge, the first study showing a clear (i.e., causal) role of Dio3 in the circannual rhythms in mammals. Having established a clear component of the circannual timing and a clean approach to address causality, this study could serve as a blueprint to decipher other components of the timing mechanism. It could also help to enlighten the elusive nature of the upstream regulators, in particular, on how the integration of day length takes place, maybe within the components in the Pars tuberalis, and the regulation of tanycytes.

Comments on revisions:

The authors have provided an improved version of the manuscript, particularly clarifying the methodology for their CRISPR manipulations. I am satisfied with their response and commend the authors for their work.

<https://doi.org/10.7554/eLife.106383.2.sa1>

Author response:

The following is the authors' response to the previous reviews.

Reviewer #1 (Public review):

Circannual timing is a phylogenetically widespread phenomenon in long-lived organisms and is central to the seasonal regulation of reproduction, hibernation, migration, fur color changes, body weight, and fat deposition in response to photoperiodic changes. Photoperiodic control of thyroid hormone T3 levels in the hypothalamus dictates this timing. However, the mechanisms that regulate these changes are not fully understood. The study by Stewart et al. reports that hypothalamic iodothyronine deiodinase 3 (Dio3), the major inactivator of the biologically active thyroid hormone T3, plays a critical role in circannual timing in the Djungarian hamster. Overall, the study yields important results for the field and is well-conducted, with the exception of the CRISPR/Cas9 manipulation.

We appreciate the positive and supportive comment from the Reviewer. We have clarified the oversight in the Crispr/Cas9 data representation below. Our correction should alleviate any concern raised.

Figure 1 lays the foundation for examining circannual timing by establishing the timing of induction, maintenance, and recovery phases of the circannual timer upon exposure of hamsters to short photoperiod (SP) by monitoring morphological and physiological markers. Measures of pelage color, torpor, body mass, plasma glucose, etc, established that the initiation phase occurred by weeks 4-8 in SP, the maintenance by weeks 12-20, and the recovery after week 20, where all morphological and physiological changes started to reverse back to long photoperiod phenotypes.

The statistical analyses look fine, and the results are unambiguous.

We thank the Reviewer for recognizing our attempts to highlight the phenomenon of circannual interval timing.

Their representation could, however, be improved. In Figures 1d and 1e, two different measures are plotted on each graph and differentiated by dots and upward or downward arrowheads. The plots are so small, though, that distinguishing between the direction of the arrows is difficult. Some color coding would make it more reader-friendly. The same comment applies to Figure S4.

We have increased the panel size for Figure 1d and 1e. We have also changed the colour of the graphs in Figure 1d and 1e to facilitate the differentiation of the two dependent variables. For the circos plots, we attempted different ways to represent the data. We have opted to keep the figures in their current stage. The overall aim is to provide a ‘gestalt’ view of the timing of changes in transcript expression and highlighted only a few key genes. The whole dataset is provided in the supplementary materials for Reviewer/Reader interrogation.

The authors went on to profile the transcriptome of the mediobasal and dorsomedial hypothalamus, paraventricular nucleus, and pituitary gland (all known to be involved in seasonal timing) every 4 weeks over the different phases of the circannual interval timer. A number of transcripts displaying seasonal rhythms in expression levels in each of the investigated structures were identified, including transcripts whose expression peaks during each phase. This included two genes of particular interest due to their known modulation of expression in response to photoperiod, Dio3 and Sst, found among the transcripts upregulated during the induction and maintenance phases, respectively. The experiments are technically sound and properly analyzed, revealing interesting candidates. Again, my main issues lie with the representation in the figure. In particular, the authors should clarify what the heatmaps on the right of Figures 1f and 1g represent. I suspect they are simply heatmaps of averaged expression of all genes within a defined category, but a description is missing in the legend, as well as a scale for color coding near the figure.

We have clarified the heatmap and density maps in the Figure legend. We apologise for the lack of information to describe the figure panels. (see lines 644-648)

Figure 2 reveals that SP-programmed body mass loss is correlated to increased Dio3-dependent somatostatin (Sst) expression. First, to distinguish whether the body mass loss was controlled by rheostatic mechanisms and not just acute homeostatic changes in energy balance, experiments from hamsters fed ad lib or experiencing an acute food restriction in both LP and SP were tested. Unlike plasma insulin, food restriction had no additional effect on SP-driven epididymal fat mass loss (Figure S7). This clearly establishes a rheostatic control of body mass loss across weeks in SP conditions. Importantly, Sst expression in the mediobasal hypothalamus increased in both ad lib fed or restriction fed SP hamsters and this increase in expression could be reduced by a single subcutaneous injection of active T3, clearly suggesting that increase in Sst expression in SP is due to a decrease of active T3 likely via Dio3 increase in expression in the hypothalamus. The results are unambiguous

We thank the Reviewer for the supportive and affirmative feedback.

Figure 3 provides a functional test of Dio3's role in the circannual timer. Mediobasal hypothalamic injections of CRISPR-Cas9 lentiviral vectors expressing two guide RNAs targeting the hamster Dio3 led to a significant reduction in the interval between induction and recovery phases seen in SP as measured by body mass, and diminished the extent of pelage color change by weeks 15-20. In addition, hamsters that failed to respond to SP exposure by decreasing their body mass also had undetectable Dio3 expression in the mediobasal hypothalamus. Together, these data provide strong evidence that Dio3 functions in the circannual timer. I noted, however, a few problems in the way the CRISPR modification of Dio3 in the mediobasal hypothalamus was reported in Figure S8. One is in Figure S8b, where the PAM sites are reported to be 9bp and 11bp downstream of sgRNA1 and sgRNA2, respectively. Is this really the case? If so, I would have expected the experiment to fail to show any effect as PAM sites need to immediately follow the target genomic sequence recognized by the sgRNA for Cas9 to induce a DNA

double-stranded break. It seems that each guide contains a 3' NGG sequence that is currently underlined as part of sgRNAs in both Fig S8b and in the method section. If this is not a mistake in reporting the experimental design, I believe that the design is less than optimal and the efficiencies of sgRNAs are rather low, if at all functional.

We apologize for the oversight and indeed the reporting in Figure S8b was a mistake. The PAM site previously indicated was the 'secondary PAM site' (which as the Reviewer notes would likely have low efficiency). The PAM site is described within the gRNA in the figure. We use Adobe Illustrator to generate figures, and during the editing process, the layer for PAM text was accidentally moved 'back' to a lower level. The oversight was not rectified before submission. We apologise for this unreservedly. The PAM site text has been moved forward, to highlight the location of the primary site (ie immediately following gRNA) and labelled the gRNA and PAM site in the 'Target region'. The secondary PAM site text was removed to eliminate any confusion.

The authors report efficiencies around 60% (line 325), but how these were obtained is not specified.

The efficiency provided are based on bioinformatic analyses and not in vivo assays. To reduce any confusion, we have removed the text. The gRNA were clearly effective to induce mutations based on the sequencing analyses.

Another unclear point is the degree to which the mediobasal hypothalamus was actually mutated. Only one mutated (truncated) sequence in Figure S8c is reported, but I would have expected a range of mutations in different cells of the tissue of interest.

The tissue punch would include multiple different cells (e.g., neuronal, glial, etc). We agree with the Reviewer that genomic samples from different cells would be included in the sequencing analyses. Given the large mutation in the target region, the gRNA was effective. We have only shown one representative sequence. If the Reviewer would like to see all mutations, we can easily show the other samples.

Although the authors clearly find a phenotypic effect with their CRISPR manipulation, I suspect that they may have uncovered greater effects with better sgRNA design. These points need some clarification. I would also argue that repeating this experiment with properly designed sgRNAs would provide much stronger support for causally linking Dio3 in circannual timing.

The gRNA was designed using the Gold-standard approach – ChopChop [citation Labon et al., 2019]. If the Reviewer's concern re design is due to the comment above re PAM site; this issue was clarified and there are no concerns for the gRNA design. The major challenge with the Dio3 gene (single exon) with a very short sequence length (approx.. 412bp). There is limited scope within this sequence length to generate gRNA.

A proposed schematic model for mechanisms of circannual interval timing is presented in Figure S9. I think this represents a nice summary of the findings put in a broader context and should be presented as a main figure in the manuscript itself rather than being relayed in supplementary materials.

We agree with the Reviewer position and moved the figure to the main manuscript. The figure is now Figure 4.

Reviewer #2 (Public review):

Several animals and plants adjust their physiology and behavior to seasons. These changes are timed to precede the seasonal transitions, maximizing chances of survival and reproduction. The molecular mechanisms used for this process are still unclear. Studies in mammals and birds have shown that the expression of deiodinase type-1, 2, and 3 (Dio1, 2, 3) in the hypothalamus spikes right before the transition to winter phenotypes. Yet, whether this change is required or an unrelated product of the seasonal changes has not been shown, particularly because of the genetic intractability of the animal models used to study seasonality. Here, the authors show for the first time a direct link between Dio3 expression and the modulation of circannual rhythms.

We appreciate the clear synthesis and support for the manuscript.

Strengths:

The work is concise and presents the data in a clear manner. The data is, for the most part, solid and supports the author's main claims. The use of CRISPR is a clear advancement in the field. This is, to my knowledge, the first study showing a clear (i.e., causal) role of Dio3 in the circannual rhythms in mammals. Having established a clear component of the circannual timing and a clean approach to address causality, this study could serve as a blueprint to decipher other components of the timing mechanism. It could also help to enlighten the elusive nature of the upstream regulators, in particular, on how the integration of day length takes place, maybe within the components in the Pars tuberalis, and the regulation of tanycytes.

We thank the Reviewer for this positive summary.

Weaknesses:

Due to the nature of the CRISPR manipulation, the low N number is a clear weakness. This is compensated by the fact that the phenotypes shown here are strong enough. Also, this is the only causal evidence of Dio3's role; thus, additional evidence would have significantly strengthened the author's claims. The use of the non-responsive population of hamsters also helps, but it falls within the realm of correlations.

We would also like to remind the Reviewer that one Crispr-Cas9 Dio3^{cc} treated hamster did not show any mutation in the genome. This hamster was observed to have a change in body mass and pelage colour like controls. This animal provides another positive control.

We also conducted a statistical power analysis to examine whether n=3 is sufficient for the Dio3^{cc} treatment group. Using the appropriate expected difference in means and standard deviations for an alpha of 0.05; we regularly observed beta >0.8 across the dependent variables.

Additionally, the consequences of the mutations generated by CRISPR are not detailed; it is not clear if the mutations affect the expression of Dio3 or generate a truncation or deletion, resulting in a shorter protein.

We agree with the Reviewer that transcript and protein assays would strengthen the genome mutation data. Due to the small brain region under investigation, we are limited in the amount of biological material to extract. Dio3 is an intronless gene and very short – approximately 412 base pairs in length. We opted to maximize resources into sequencing the gene as the confirmation of genetic mutation is paramount. Given the large size of the

mutation in the treated hamsters, there would be no amplification of transcript or protein translated.

Reviewer #3 (Public review):

The authors investigated SP-induced physiological and molecular changes in Djungarian hamsters and the endogenous recovery from it after circa half a year. The study aimed to elucidate the intrinsic mechanism and included nice experiments to distinguish between rheostatic effects on energy state and homeostatic cues driven by an interval timer. It also aimed to elucidate the role of Dio3 by introducing a targeted mutation in the MBH by ICV. The experiments and analyses are sound, and the amount of work is impressive. The impact of this study on the field of seasonal chronobiology is probably high.

We thank the Reviewer for their positive comments and support for our work.

Even though the general conclusions are well-founded, I have fundamental criticism concerning 3 points, which I recommend revising:

(1) The authors talk about a circannual interval timer, but this is no circannual timer. This is a circasemiannual timer. It is important that the authors use precise wording throughout the manuscript.

We agree with the Reviewer that the change in physiology and behaviour does not approximate a full year (e.g. annual) and only a half of the year. We opted to use circannual timer as this term is established in the field (see doi: 10.1177/0748730404266626; doi: 10.1098/rstb.2007.2143). We cannot identify any publication that has used the term ‘semiannual timer’. We do not feel this manuscript is the appropriate time to introduce a new term to the field; we will endeavour to push the field to consider the use of ‘semiannual timer’. A Review or Opinion paper is best place for this discussion. We hope the Reviewer will understand our position.

(2) The authors put their results in the context of clocks. For example, line 180/181 seasonal clock. But they have described and investigated an interval timer. A clock must be able to complete a full cycle endogenously (and ideally repeatedly) and not only half of it. In contrast, a timer steers a duration. Thus, it is well possible that a circannual clock mechanism and this circa-semiannual timer of photoperiodic species are 2 completely different mechanisms. The argumentation should be changed accordingly.

We agree with the Reviewers definitions of circannual ‘clock’ and ‘timer’. We were careful to distinguish between the two concepts early in the manuscript (lines 41-46). We have added italics to emphasis the different terms. The use of seasonal clock on line 180/191 was imprecise and we appreciate the Reviewer highlighting our oversight and the text was revised. We have also revised the Abstract accordingly.

(3) The authors chose as animal model the Djungarian hamster, which is a predominantly photoperiodic species and not a circannual species. A photoperiodic species has no circannual clock. That is another reason why it is difficult to draw conclusions from the experiment for circannual clocks. However, the Djungarian hamster is kind of "indifferent" concerning its seasonal timing, since a small fraction of them are indeed able to cycle (Anchordoquy HC, Lynch GR (2000), Evidence of an annual rhythm in a small proportion of Siberian hamsters exposed to chronic short days. J Biol Rhythms 15:122-125.). Nevertheless, the proportion is too small to suggest that the findings in the current study might reflect part of the circannual timing. Therefore, the authors should make a clear distinction between timers and clocks, as well as between circa-annual and circa-semiannual durations/periods.

This comment is not clear to us. The Reviewer states the hamsters are not a circannual species, but then highlight one study that shows circannual rhythmicity. We agree that circannual rhythmicity in Djungarian hamsters is dependent on the physiological process under investigation (e.g. body mass versus reproduction) and that photoperiodic response system either dampen or mask robust cycles. We have corrected the text oversight highlighted above and the manuscript is focused on interval timers. We have kept the term circannual over semicircannual due to the prior use in the scientific literature.

Reviewing Editor Comments:

The detailed suggestions of the reviewers are outlined below (or above in case of reviewer 1). In light of the criticism, we ask the authors to especially pay attention to the comments on the Cas9/Crisp experiment, raised by Reviewers 1 and 2. As currently described, there are serious questions on the design of the sgRNAs, and also missing critical methodological details. If the latter are diligently taken care of, they may resolve the questions on the sgRNA design. Please also reconsider the wording along the suggestions of Reviewer 3.

We appreciate the Editors time and support for the manuscript. We have clarified and corrected our oversight for the PAM site. This correction confirms the strength of the Cas9 gRNA used in the study. The correction should remove all concerns. We have also considered using semicircannual in the text. As there is existing scientific literature using circannual interval timer, and there is no publication to our knowledge for using 'semicircannual'; we have opted to keep with the current approach and use circannual. We feel a subsequent Opinion paper is more suitable to introduce a new term.

Reviewer #2 (Recommendations for the authors):

First, I want to commend the authors for their work. It is a clear advancement for our field. Below are a couple of comments and suggestions I have:

we thank the Review for the positive comment and support. We have endeavoured to incorporate their suggested improvements to the manuscript.

(1) Looking at the results of Figure 1A and Figure S8, the control in S8 showed a lower pelage color score as compared to the hamsters in 1A. Is this a byproduct of the ICV injection?

The difference between Figure 1 and 3 is likely due to the smaller sample sizes. The controls in Figure 1 had a higher proportion of hamsters show complete white fur (score =3) at 1618 weeks compared to controls in Figure 3. It is possible, although unlikely that the ICV injection would reduce the development of winter phenotype. There was no substance in the ICV injection that would impact the prolactin signalling pathway. Our perspective is that the difference between the two figures is due to the different sampling population. Overall, the timing of the change in pelage colour is the same between the figures and suggest that the mechanisms of interval timer were unaffected.

(2) Is there a particular reason why the pelage color for the CRISPR mutants is relegated to the supplemental information? In my opinion, this is also important, even though the results might be difficult to explain. Additionally, did the authors check for food intake and adipose mass in these animals?

We agree with the Reviewer the pelage change is very interesting. We decided to have Figure 3 focus on body mass. The rationale was due to the robust nature of the data collection from

Crispr-cas9 study (Fig.3b), in addition to the non-responsive hamsters (Fig.3e). We disagree that the data patterns are hard to explain, as pelage changes was similar to the photoperiodic induced change in body mass. No differences were observed for food intake or adipose tissue. We have added this information in the text (see lines 162-163).

(3) I might have missed it, but did the authors check for the expression of Dio3 on the CRISPR mutants? Does the deletion cause reduced expression or any other mRNA effect, such as those resulting in the truncation of a protein?

Due to the limited biological material extracted from the anatomical punches, we decided to focus on genomic mutations. Dio3 has a very short sequence length and the size of the mutations identified indicate that no RNA could be transcribed.

(4) Could the authors clarify which reference genome or partial CDS (i.e., accession numbers) they used to align the gRNA? Did they use the SSSS strain or the Psun_Stras_1 isolate?

The gRNAs were designed using the online tool CHOPCHOP, using the *Mus musculus* Dio3 gene. The generated gRNAs were subsequently aligned via blast with the *Phodopus sungorus* Dio3 partial cds (GenBank: MF662622.1), to ensure alignment with the species. We are confident that the gRNA designed align 100% in hamsters. Furthermore, we conducted BLAST to ensure there were no off-targets. The only gene identified in the BLAST was the rodent (i.e. hamster, mouse) Dio3 sequence.

(5) Figure 3b. I do agree with the authors in pointing out that the decrease in body mass is occurring earlier in Dio3wt hamsters; however, the shape of the body mass dynamic is also different. Do the authors have any comments on the possible role of Dio3 in the process of exist of overwintering?

This is a very interesting question. We do not have the data to evaluate the role of Dio3 for overwintering. We argue that disruption in Dio3 reduced the circannual interval period. For this interpretation, yes, Dio3 is necessary for overwintering. However, we would need to show the sufficiency of Dio3 to induce the winter phenotype in hamsters housed in long photoperiod. At this time, we do not have the technical ability to conduct this experiment.

(6) In Figure 3d, the Dio3wt group does not show any dispersion. Is this correct? If that's true, and no dispersion is observed, no normality can be assumed, and a t-test can't be performed (Line 692).The Mann-Whitney test might be better suited.

We conducted a Welch's t-test to compare the difference in body mass period. We used the Welch's test as the variance were not equal; Mann-Whitney test is best for skewed distributions. To clarify the test used, we have added 'Welch's test' to the Figure legend.

(9) Figure 1 h. It might be convenient to add the words "Induction", "maintenance", and "recovery" over each respective line on the polar graph for easier reading.

We have added the text as suggested by the Reviewer.

Reviewer #3 (Recommendations for the authors):

(1) Figure 1: Please enlarge all partial graphics at least to the size of Figure 2. In the print version, labels are barely readable

we have increased the panels in Figure 1 and 3 by 20% to accommodate the Reviewers suggestion.

(2) *Legend Figure 2: Add that the food restriction was 16h.*

We have added 16h to the text.

(3) *Figure 3b: enlarge font size. In the legend: Dio3cc hamsters delayed.... The delay might have been a week or so, but not more (and even that is unclear since the rise in body mass in that week seems to be rather a disturbance of the curve). Thus 'delay' might not be the most appropriate wording. Instead, the initial decline is slower, but both started at nearly the same week (=> no delay). Minimum body mass is reached at the identical week as in wt (=> no delay). Also, the increase started at the same week but was much faster in Dio3cc than in wt. Figure 3c: How can there be a period when there is no repeated cycle (rhythm)? This is rather a duration. Moreover, according to the displayed data, I am wondering which start point and which endpoint is used. The first and last values are the highest of the graph, but have they been the maximum? Especially for Dio3wt, it can be assumed that animals haven't reached the maximum at the end of the graph.*

We have increased the font size in Figure 3b. We have changed 'delayed' to 'slower' in the text. Period analyses, such as the Lomb-Scargle measure the duration of a cycle (and multiple cycles). The start point and end point used in the analyses were the initial data collection date (week 0) and the final data collection date (week 32). The Lomb-Scargle analyses determines the duration of the period that occurs within these phases of the cycle. We believe the period analyses conducted by the Lomb-Scargle is the most suitable for the scientific question.

(4) *Figure S9: This is a very nice graph and summarises your main results. It should appear in the main manuscript and not in the supplements.*

We appreciate the positive comment and suggestion. We agree with the Reviewer and have move the graph to the main figure. The revised manuscript indicates the graph as Figure 4.

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